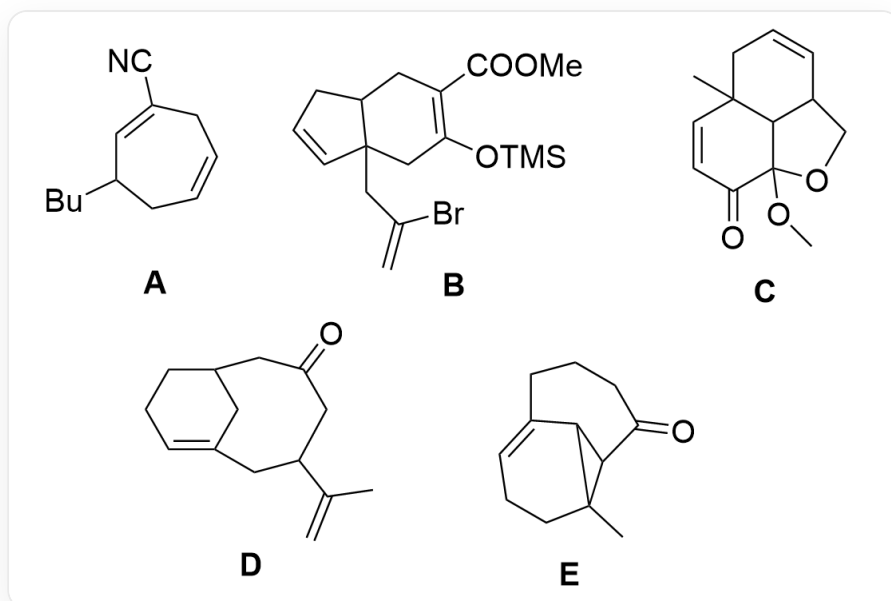


Question



This figure illustrates 5 organic structural formulas. **A** is N#CC1=CC(CCCC)CC=CC1; **B** is C=C(Br)CC12C=CCC1CC(C(OC)=O)=C(O[Si](C)(C)C)C2; **C** is CC1(C=C2)CC=CC3C1C(OC3)(OC)C2=O; **D** is O=C(CC(C(C)=C)C1)CC2CC1=CCC2; **E** is CC12CCC=C(CCC3)C1C2C3=O.

The structures in the above figure are all generated through Cope rearrangement or O-Cope rearrangement; regarding the precursor structural formulas before they undergo rearrangement, the following statements are made:

1. The rearrangement precursor corresponding to **A** contains a six-membered ring.
2. The rearrangement precursor corresponding to **B** does not contain a five-membered ring.
3. The rearrangement precursor corresponding to **C** contains only two six-membered rings and one five-membered ring.
4. The rearrangement precursors corresponding to **D**, **E** both contain a hydroxyl group.
5. The rearrangement precursor corresponding to **E** contains a three-membered ring.

6. The rearrangement precursor corresponding to **D** reacts with $\text{O}_3, \text{Zn}/\text{MeOH}$ to obtain a new structure containing two carbonyl groups.

Among the following options, the one that contains the largest number of correct statement numbers is:

A. All other options are incorrect

B. 1, 3

C. 1, 4

D. 2, 5

E. 3, 5

F. 4, 5

G. 5, 6

H. 2, 6

I. 1, 3, 4

J. 1, 4, 5

K. 1, 2, 6

L. 2, 4, 5

M. 2, 4, 6

N. 3, 5, 6

O. 4, 5, 6

P. 1, 3, 4, 5

Q. 1, 4, 5, 6

R. 1, 2, 3, 4

S. 1, 3, 5, 6

T. 3, 4, 5, 6

U. 1, 2, 4, 5, 6

V. 2, 3, 4, 5, 6

W. 1, 2, 3, 4, 5, 6

X. 1, 3, 4, 6

Y. 1, 2, 4, 5

Answer

Correct Answer: **O**

Detailed Explanation

The general formula for the Cope rearrangement is:

1,5-dienes undergo a $[3,3]\sigma$ sigmatropic rearrangement under thermal conditions to yield a new 1,5-diene.

CHECKPOINT

1 PTS

Cope rearrangement is a $[3,3]\sigma$ sigmatropic rearrangement that 1,5-dienes undergo under thermal conditions to yield a new 1,5-diene.

This reaction forms the C1 – C6 bond, breaks the C3 – C4 bond, and simultaneously migrates the double bond from the 1,5-position to the 3,4-position.

CHECKPOINT

1 PTS

Formation of C1 – C6 bond, breaking of C3 – C4 bond

The characteristic of the O-Cope rearrangement compared to the Cope rearrangement is that the substrate contains a hydroxyl group at the 1 or 4 position, and after undergoing a $[3,3]\sigma$ sigmatropic rearrangement, it will produce an enol structure at the 1 or 4 position, thereby converting to a carbonyl group at the 1 or 4 position.

CHECKPOINT

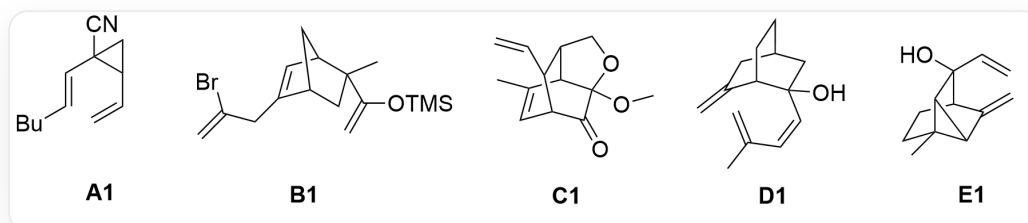
1 PTS

O-Cope rearrangement produces an enol structure at the 1 or 4 position, thereby converting to a carbonyl group at the 1 or 4 position

Therefore, the steps for retro-analyzing the Cope rearrangement and O-Cope rearrangement are:

1. Convert the carbonyl group to an enol structure;
2. Find the 1,5-diene structure and number it;
3. Break the C1 – C6 bond, form the C3 – C4 bond, and migrate the double bond.

Therefore, according to the above steps, the precursors corresponding to **A** – **E** can be deduced as shown in the following figure:



A1 is C=CC1C(C1)(C#N)/C=C/CCCC ; **B1** is C[C@]1(C(O[Si](C)(C)C)=C)[C@H](C2)C=C(CC(Br)=C)[C@H]2C1 ; **C1** is C=CC1C2C(C(C)=CC1C3=O)C3(OC)OC2 ; **D1** is C=C1C[C@@H]2C[C@@](/C=C\C(C)=C)(O)[C@H]1CC2 ; **E1** is CC1(C2C1C3(C=C)O)CCC3C2=C °

The precursor **A1** corresponding to **A** is C=CC1C(C1)(C#N)/C=C/CCCC.

CHECKPOINT

2 PTS

The precursor **A1** corresponding to **A** is C=CC1C(C1)(C#N)/C=C/CCCC.

The precursor **B1** corresponding to **B** is C[C@]1(C(O[Si](C)(C)C)=C)[C@H](C2)C=C(CC(Br)=C)[C@H]2C1.

CHECKPOINT

2 PTS

The precursor **B1** corresponding to **B** is C[C@]1(C(O[Si](C)(C)C)=C)[C@H](C2)C=C(CC(Br)=C)[C@H]2C1.

The precursor **C1** corresponding to **C** is C=CC1C2C(C(C)=CC1C3=O)C3(OC)OC2.

CHECKPOINT

2 PTS

The precursor **C1** corresponding to **C** is C=CC1C2C(C(C)=CC1C3=O)C3(OC)OC2.

The precursor **D1** corresponding to **D** is C=C1C[C@@H]2C[C@@]/(C=C\C(C)=C)(O)[C@H]1CC2.

CHECKPOINT

2 PTS

The precursor **D1** corresponding to **D** is C=C1C[C@@H]2C[C@@]/(C=C\C(C)=C)(O)[C@H]1CC2.

The precursor **E1** corresponding to **E** is CC1(C2C1C3(C=C)O)CCC3C2=C.

CHECKPOINT

2 PTS

The precursor **E1** corresponding to **E** is CC1(C2C1C3(C=C)O)CCC3C2=C.

D1 is cleaved by ozonolysis of the double bond, and because it contains a terminal olefin, it will only produce two carbonyl groups.

According to the structural formula, only statements 4, 5, and 6 are correct.

CHECKPOINT

1 PTS

Statements 4, 5, and 6 are correct